Clearity

Cashier video review with timestamped evidence

ForteBank Datathon 26 prototype

Local inference. Human review.

github.com/Asylkenei/clearity

The manual review problem

Brief actions demand attention

Silent footage lacks transaction context.
Occlusion and camera angle limit what is visible.

10–145 s

jury video duration

A visual signal requires human interpretation.

≤ 5 min

required processing limit

Input format: MOV, approximately 29 MB per video

A review journey built around evidence

Upload MOV → Job status → Binary result → Human review

Evidence attached to candidate events

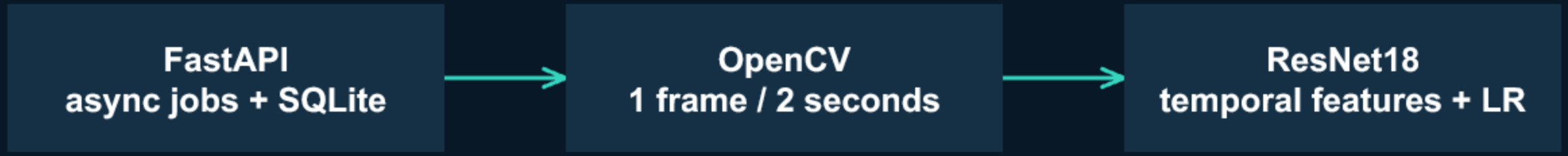
Timestamps and 1–3 JPEG frames per event
Explanation with Kazakhstan regulatory context

POST /api/v1/analyze **GET /api/v1/jobs/{job_id}**

Binary labels support screening. Semantic violation subtypes remain unvalidated.

Inference inside the service

HOSTED BASELINE



OPTIONAL LOCAL EXPERIMENT

Qwen3-VL-2B GGUF through llama.cpp
3 frames per 20-second window. Separate from the free hosted pipeline.

X-API-Key authentication. No external AI API. No face identification.
Uploaded video copies are deleted after processing.

A transparent experimental baseline

40
unique videos
42 labeled records

24 / 8 / 8
train / validation / test

50%
held-out accuracy
on 8 videos

Precision 50% **Recall 50%** **F1 0.50**

Regulatory context: Kazakhstan National Bank Resolution No. 231, 29 Nov 2019.
Verified event-to-clause mapping remains unfinished.

Saved baseline experiment. Production reliability and camera-angle robustness are unproven.

Demo storyboard and verification status

01 Upload

MOV submission
Screenshot placeholder

02 Processing

Asynchronous job state
Screenshot placeholder

03 Review

Prediction + evidence
Screenshot placeholder

63.6 s hosted end-to-end test

145-second / 29 MiB synthetic MOV. Warm service, one run.

Repository: github.com/Asylkenei/clearity

Live API: clearity-3qoy.onrender.com. Cold-start timing remains unverified.

Next steps toward reliable review

More independent data and camera-angle tests

Event annotations and better temporal localization

Expert review of regulatory references

Hosted benchmarks, cold starts and retention checks

Evidence for human judgment